

Midas_V2 — Scenarios Documentation (v0.3.2-scenarios)

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This document describes scenarios A–E as they currently exist, high-level descriptions, how we tested them, results observed, why Scenario A can be deceptive, and the exact run commands to reproduce. An appendix provides parameter definitions.

1) Scenario Definitions (Current Configuration)

A — Loose exploratory profile. Minimal safeguards, MACD off, gate=0, wider SL. Trades very frequently and may look good on strong-trend days but is risky on choppy days.

Parameter	Value
min_pm_vol	0
dip_reclaim	False
tp_pct	1.2
sl_pct	2.7
riseBars	2
min_dip_pct	2
macd_confirm	True
ema_confirm	True
vwap_confirm	True
ema_period	9
gate_minutes	0
reclaim_pmh	False
reclaim_ref	ema
min_reclaim_pct	0.5

B — Safe baseline. Aligned with Cameron-style rules (MACD on, EMA/VWAP confirm, gate=10, TP/SL ~2/2.5, min_pm_vol=30k). Designed to be reproducible and consistent.

Parameter	Value
min_pm_vol	30000
dip_reclaim	False
tp_pct	2.0
sl_pct	2.5
riseBars	2
min_dip_pct	2
macd_confirm	True
ema_confirm	True
vwap_confirm	True
ema_period	9

gate_minutes	10
reclaim_pmh	False
reclaim_ref	ema
min_reclaim_pct	0.5
min_price	1
max_price	20
min_gap_pct	10
max_gap_pct	40

C — Variant of A. Similar looseness with slight differences (e.g., VWAP confirm off, MACD off). Research profile, not recommended for baseline trading.

Parameter	Value
min_pm_vol	0
dip_reclaim	False
tp_pct	1.2
sl_pct	2.7
riseBars	2
min_dip_pct	2
macd_confirm	False
ema_confirm	True
vwap_confirm	False
ema_period	9
gate_minutes	0
reclaim_pmh	False
reclaim_ref	ema
min_reclaim_pct	0.5

D — Strict profile. Now updated to MACD on, gate=5, SL=2.5, min_pm_vol=30k, with EMA/VWAP confirm and price/gap bands. Trades less often, with stronger filters.

Parameter	Value
min_pm_vol	0
dip_reclaim	False
tp_pct	2.0
sl_pct	3.5
riseBars	2
min_dip_pct	2
macd_confirm	False
ema_confirm	True

vwap_confirm	True
ema_period	9
gate_minutes	0
reclaim_pmh	False
reclaim_ref	ema
min_reclaim_pct	0.5

E — Dip-reclaim strategy. Looks for dips to VWAP/EMA with MACD rising. Gate=10, PM vol=30k. Adds profit-factor potential when reclaim plays set up.

Parameter	Value
min_pm_vol	30000
dip_reclaim	True
tp_pct	2.0
sl_pct	2.5
riseBars	2
min_dip_pct	2.0
macd_confirm	True
ema_confirm	True
vwap_confirm	True
ema_period	9
gate_minutes	10
reclaim_pmh	False
reclaim_ref	vwap
min_reclaim_pct	0.5

B_safe — No high-level description provided.

Parameter	Value
min_pm_vol	30000
dip_reclaim	False
tp_pct	2.0
sl_pct	2.5
riseBars	2
min_dip_pct	2
macd_confirm	True
ema_confirm	True
vwap_confirm	True
ema_period	9
gate_minutes	10

reclaim_pmh	False
reclaim_ref	ema
min_reclaim_pct	0.5

D_backup — No high-level description provided.

Parameter	Value
min_pm_vol	0
dip_reclaim	False
tp_pct	2.0
sl_pct	3.5
riseBars	2
min_dip_pct	2
macd_confirm	False
ema_confirm	True
vwap_confirm	True
ema_period	9
gate_minutes	0
reclaim_pmh	False
reclaim_ref	ema
min_reclaim_pct	0.5

B_backup — No high-level description provided.

Parameter	Value
min_pm_vol	0
dip_reclaim	False
tp_pct	1.2
sl_pct	2.7
riseBars	2
min_dip_pct	2
macd_confirm	False
ema_confirm	True
vwap_confirm	True
ema_period	9
gate_minutes	0
reclaim_pmh	False
reclaim_ref	ema
min_reclaim_pct	0.5

2) How We Tested the Scenarios

We ran single-day backtests and a day summary on 2025-08-05, using the project's built-in runners. The flow mirrors your USER_GUIDE:

- Fetch top gappers for the day (internal script): scripts/topgappers.py
- Write the universe file (auto): data/samples/universe_sample.txt
- Fetch minute data (Polygon): scripts/fetch_minutes_polygon.py --date 2025-08-05 --session rth
- Run backtest per scenario: python -m midas_v2.cli backtest ... (called by scripts/run_day_simple.py)
- Results written to: out///results_.csv
- Summarize: scripts/summarize_results.py --date YYYY-MM-DD

Single-day commands used:

```
python scripts\run_day_simple.py --date 2025-08-05 --scenario D
python scripts\run_day_simple.py --date 2025-08-05 --scenario B
python scripts\summarize_results.py --date 2025-08-05
```

3) Results We Saw (2025-08-05) and What They Mean

Scenario	Result
A	TP=29 SL=10 Win%=74.36
B	TP=1 SL=0 Win%=100.0
C	TP=29 SL=10 Win%=74.36
D	TP=1 SL=0 Win%=100.0
E2	TP=2 SL=0 Win%=100.0
E	TP=0 SL=0 Win%=0
E_cfg2	TP=0 SL=0 Win%=0
E_cfg	TP=0 SL=0 Win%=0
E_debug	TP=0 SL=0 Win%=0
E_vwapoff	TP=0 SL=0 Win%=0

Interpretation:

- B vs D: Both captured the clean STTK breakout (TP ~2.04%). D is stricter and should trade less over multiple days while maintaining or slightly improving win rate.
- A/C: Looser profiles with many more trades (39 trades, 29 TP, 10 SL) with ~74% WR on this date—useful for research, but risky in practice.
- E2: Dip-reclaim variant, fired twice and won both. Shows promise, requires more dates to validate.
- E family (others): No trades on this date. Normal when reclaim criteria don't occur.

4) Why Scenario A's Results Can Be Deceptive

Scenario A looked strong on 2025-08-05, showing 29 wins out of 39 trades (~74% WR). But this performance can be misleading:

- Overtrading: A fired repeatedly on SMXT, taking dozens of trades on one ticker. On a trending day this looks good, but it exposes you to clustering risk.
- Survivorship bias: A benefits from strong trend days. On choppy days, those same loose filters would produce many stop-outs.
- Single-ticker concentration: Nearly all trades came from one symbol, creating overexposure.

- Hidden risk: Multiple stop-outs can cluster if the ticker reverses. This is why Ross and our stricter configs avoid overtrading weak gappers.

By contrast, Scenarios B and D only allowed the cleanest trade, keeping risk contained. This safeguard makes their results more reliable across market conditions, even if they show fewer winners on strong days.

5) Reproduction Commands (Detailed)

Single day run (baseline and strict):

```
python scripts\run_day_simple.py --date 2025-08-05 --scenario B
python scripts\run_day_simple.py --date 2025-08-05 --scenario D
python scripts\summarize_results.py --date 2025-08-05
```

Optional range run:

```
python scripts\run_range_and_summarize.py --start 2025-08-05 --end 2025-08-07 --scenarios
B,D,E
```

Version tag (two-step, safe in PowerShell):

```
git add -A
git commit -m Scenarios: D strict (MACD on, gate=5, SL=2.5, 30k PM vol, price/gap bands)
git tag -a v0.3.2-scenarios -m Milestone: Scenario D strict; B baseline; E dip-reclaim
git push
git push origin v0.3.2-scenarios
```

Appendix A — Parameter Glossary

Parameter	Description
min_pm_vol	Minimum premarket volume required for the ticker to qualify.
tp_pct	Take-profit percentage target. Trade exits at this gain.
sl_pct	Stop-loss percentage. Trade exits at this loss.
rise_bars	Number of consecutive rising bars required for momentum confirmation.
min_dip_pct	Minimum dip percentage from high required to qualify for a dip reclaim.
macd_confirm	Require MACD histogram rising or MACD line > signal line.
ema_confirm	Require price above EMA and EMA trending up.
vwap_confirm	Require price above VWAP and VWAP sloping up.
ema_period	Period (bars) used for the EMA confirm.
gate_minutes	Minimum minutes after open before trades allowed (e.g., gate=10 skips first 10m).
dip_reclaim	If true, scenario waits for dip below EMA/VWAP then reclaim trigger.
reclaim_pmh	If true, scenario requires reclaim of premarket high.
reclaim_ref	Reference indicator for reclaim logic (e.g., 'ema' or 'vwap').
min_reclaim_pct	Minimum % threshold required for reclaim confirmation.
min_price	Minimum stock price allowed for scenario.
max_price	Maximum stock price allowed.
min_gap_pct	Minimum gap-up percentage required at open.
max_gap_pct	Maximum gap-up percentage allowed.